Travel Agent

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香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

JC STEM Lab of
Data Science Foundations
賽馬會數據科學基礎創科實驗室

Problem

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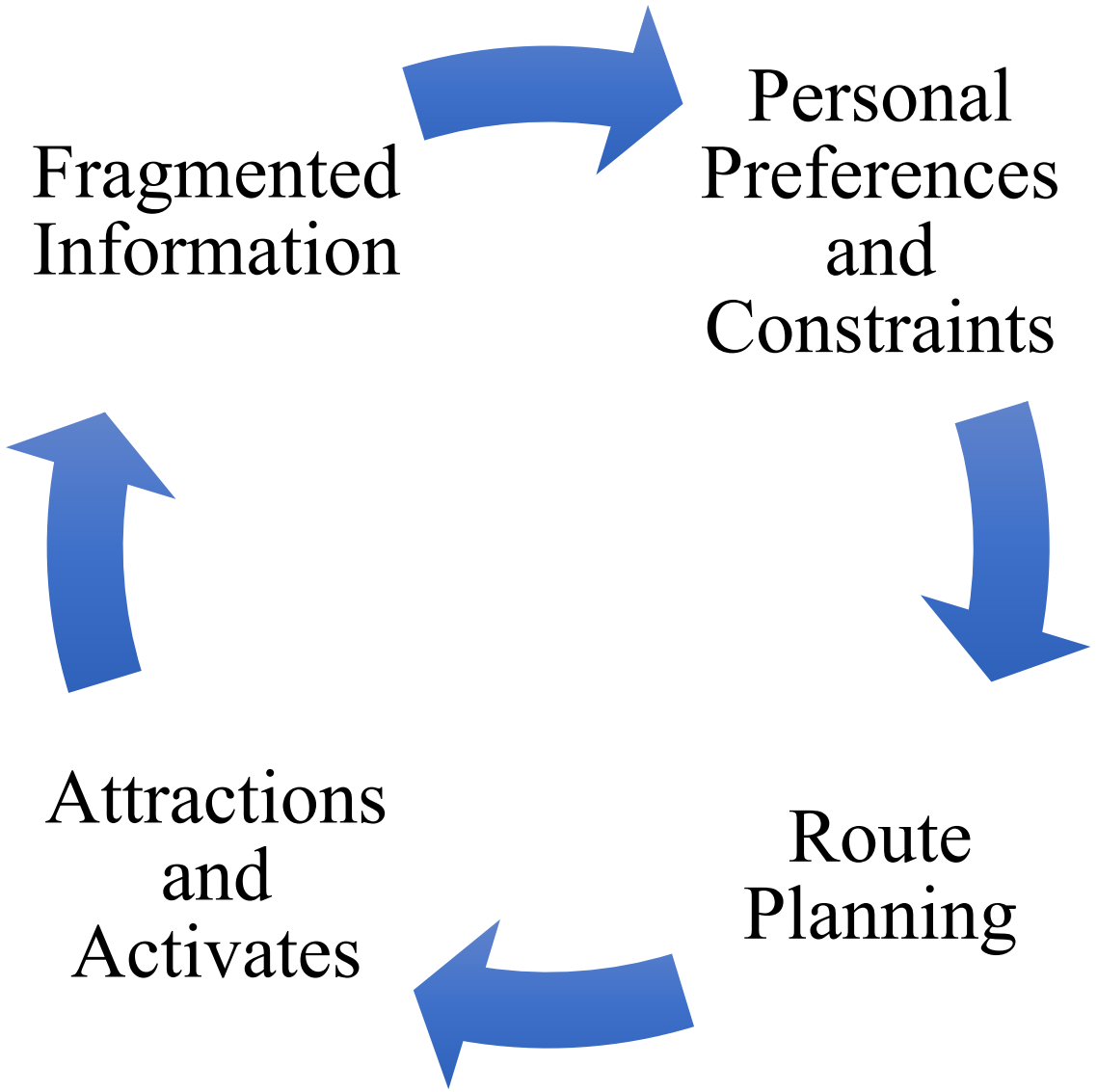
Problem:
Planning a trip is labor intensive!



We want to build an AI travel agent that can:

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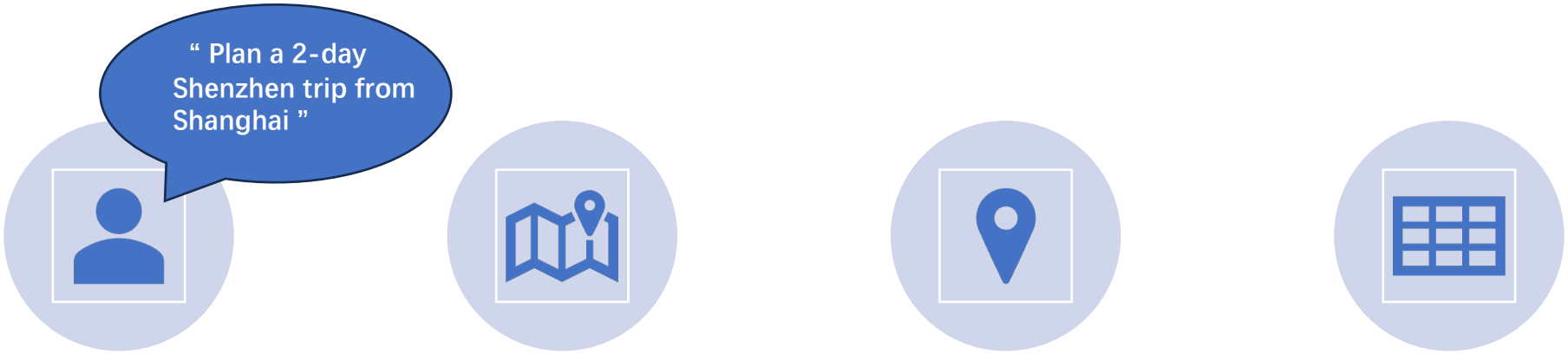

UNDERSTAND NATURAL LANGUAGE TRAVEL REQUESTS

GENERATE MULTI-DAY TRAVEL ITINERARIES

INTEGRATE EXTERNAL INFORMATION SOURCES

REASON OVER MULTIPLE CONSTRAINTS (DATE, PREFERENCES, BUDGETS)

Overview

Core Processes:

1. Intent Extraction and Clarification (Extract)
2. Search and Draft Initial Version (Search & Draft)
3. Refine the Logic of Progression (Refine)
4. Resource Enrichment and Aggregated Output (Enrich)

Overview

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Tools & APIs

1. LLM / Model API🧠:

   System's core reasoning engine that understands user requests, extracts travel preferences and generates the overall travel itinerary.

2. Amap Live Service API🗺️

   System's geographical data provider, offering real-world location information and distance verification to ensure routes are geographically feasible.

3. Serper API(Google Search Aggregation) 🔍

   A web search tool that retrieves images, travel guides, attraction details, and price information to enrich the travel itinerary.

Architecture

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Frontend:

- collect user query and selected model
- submit to FastAPI
- render clarification form when needed
- render timeline, logistics, references, and route map when itinerary is ready

Backend

- resolve model credentials from .env
- call the selected LLM using the existing compatible chat-completions client
- collect grounding context from tools
- generate and refine itinerary data and persist trips in SQLite

Planning Flow

1. user enters a free-form query
2. backend parses the request heuristically
3. backend asks the selected model to extract normalized constraints
4. backend gathers tool context:
    1. Amap for hotels, restaurants, POIs, and walking directions
    2. SerperApi as the primary search engine
5. backend runs a draft LLM pass
6. backend runs a refinement LLM pass with search-grounded context
7. backend hydrates:hotel, POIs, images, route trip visuals
8. frontend renders the final result

Limitation

- The planning process is a bit slow.
- Amap api sometimes does not map the POI into the real address and it only accept the Chinese prompt and Chinese cities.
- We do not have access to many real world information like the flight, the price of visiting attractions, the price of the hotel and the train(12306).

Ending

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